Rubric and Solution of Quiz 3 Part 2

This file contains the **rubric and solution of Quiz 3 Part 2**.

Note;- There may be multiple correct ways to solve a problem.
The document presents one representative approach, and the rubric is based on that.
If your solution follows a different valid method with proper justification, appropriate marks will be awarded.

1. (4 marks)

Claim: "Every injective function $F:X \to Y$ is also surjective."
Answer: False. Provide counterexamples and reasoning.

Finite counterexample (simple, clear):
Take $X=\{1,2\}$, $Y=\{a,b,c\}$. Define $F(1)=a$, $F(2)=b$. F is injective (distinct inputs map to distinct outputs) but not surjective because $c \notin F(X)$. Hence the claim is false.

Infinite counterexample (shows generality):
Let $X=Y=N$ and define $F(n)=n+1$. F is injective but not surjective because 1 is not in the image. So again the claim fails.

Note:- only finite counterexample is not sufficient because question have not mentioned finite. we have to give counterexample for Infinite for generality

Reasoning (why the claim is false in general):
Injectivity only guarantees different inputs map to different outputs; it does **not** guarantee every element of Y has a preimage. Only when $|X|=|Y|$ (and both finite) or F is explicitly bijective can you conclude surjectivity. Because there exist easily-constructed functions (above) that are injective but miss some $y \in Y$, the universal statement is false.

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2. (3 marks)

Let Y contain m rationals, Z contain n irrationals (so $|Y|=m$, $|Z|=n$), X arbitrary finite set of integers. For an arbitrary function $F:X \to Y$, how large must $|X|$ be to **guarantee** F is **not injective**?

Answer: at least mn + 1.

Justification (pigeonhole principle):
The codomain $Y \times Z$ has exactly $|Y \times Z|=m \cdot n$ elements. An injective function from X into $Y \times Z$ can have at most mn distinct images. If $|X| \geq mn+1$, then there are more domain elements than available distinct targets; by the pigeonhole principle two different elements of X must map to the same element of $Y \times Z$. Hence F cannot be injective. Thus any set X with mn+1 elements guarantees non-injectivity. (If $|X| \leq mn$ injectivity might still be possible; the guarantee requires at least mn + 1)

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3. (3 marks)

Refer lecture 13 page no 12

1.1.1 (4 marks)

- **4/4** → Stating False + correct example + clear reasoning.
- **3/4** → Correct answer but missing clear justification.
- **2/4** → Right idea but example or reasoning flawed.
- **0/4** → Incorrect claim or no reasoning provided.

1.1.2 (3 marks)

- **3/3:** Stating mn + 1, reasoning, and principle clearly stated.
- **2/3:** correct formula mn+1 but weak or missing justification.
- **0/4** → Incorrect claim or no reasoning provided.

1.1.3 (3 marks)

- **1.5/3:** showing forward subset proof.
- **1.5/3:** showing backward subset proof.
- **0/3** →missing forward or backward subset proof and not solved using propositional logic.

Rubric quiz 3 part 2 Q 1.2

Question 1 (4 marks)

$$F(x) = \begin{cases} x+3, & \text{if } x \text{ is odd,} \\ x-3, & \text{if } x \text{ is even.} \end{cases}$$

(a) Is F a function? (1 mark)

Each integer is either even or odd, never both. Hence, for every input x, exactly one rule applies and $F(x)$ is uniquely defined.

$$\Rightarrow \boxed{F \text{ is a well-defined function.}}$$

(b) Injectivity (1.5 marks)

Assume $F(x_1) = F(x_2)$. We consider all parity cases:

- If both x_1, x_2 are odd: $x_1 + 3 = x_2 + 3 \Rightarrow x_1 = x_2$.
- If both are even: $x_1 - 3 = x_2 - 3 \Rightarrow x_1 = x_2$.
- If one odd and one even: outputs have opposite parity, so cannot be equal.

$$\boxed{F \text{ is injective.}}$$

(c) Surjectivity (1 mark)

Let $y \in \mathbb{Z}$.

- If y is even, take $x = y - 3$ (odd). Then $F(x) = (y-3) + 3 = y$.
- If y is odd, take $x = y + 3$ (even). Then $F(x) = (y+3) - 3 = y$.

$$\boxed{F \text{ is surjective.}}$$

(d) Conclusion and Inverse (0.5 mark)

Since F is both injective and surjective, F is bijective.

Inverse:

$$F^{-1}(y) = \begin{cases} y-3, & \text{if } y \text{ is even,} \\ y+3, & \text{if } y \text{ is odd.} \end{cases}$$

$$\boxed{F \text{ is bijective.}}$$

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Solutions for Question 1.2.2 and 1.2.3

Course: Discrete Mathematics

Question 1.2.2 (Total 2.5 marks)

Statement: Let A, B be arbitrary sets. Prove or disprove:

$$A \subset B \text{ if and only if } \text{pow}(A) \subset \text{pow}(B),$$

where $\subset$ denotes a **proper subset**.

Proof

(1 mark) () **Forward Direction:** Assume $A \subset B$ (that is, A is a proper subset of B). Then:

$$\forall x(x \in A \Rightarrow x \in B), \quad \text{and} \quad \exists y(y \in B \text{ and } y \notin A).$$

Now, consider any subset S of A. Since every element of A is in B, we have $S \subseteq B$, implying that $S \in \text{pow}(B)$. Therefore:

$$\text{pow}(A) \subseteq \text{pow}(B).$$

To check for a **proper subset** relation, note that because $A \subset B$, there exists at least one element $y \in B \setminus A$. Then $\{y\} \subseteq B$ but $\{y\} \nsubseteq A$. Hence, $\{y\} \in \text{pow}(B)$ but $\{y\} \notin \text{pow}(A)$, implying:

$$\text{pow}(A) \subset \text{pow}(B).$$

(1 mark) () **Reverse Direction:** Assume $\text{pow}(A) \subset \text{pow}(B)$. Since $A \in \text{pow}(A)$, this means $A \in \text{pow}(B)$, hence $A \subseteq B$.

To verify that A is a **proper subset** of B, note that the inclusion between the power sets is proper. Therefore, there exists a subset $S \in \text{pow}(B)$ such that $S \notin \text{pow}(A)$. This means $S \subseteq B$ but $S \nsubseteq A$, so there exists $x \in S$ such that $x \notin A$. Hence, $x \in B \setminus A$, implying $A \neq B$. Thus:

$$A \subset B.$$

(0.5 mark) **Conclusion:** The statement is **true** when $\subset$ denotes a proper subset:

$$A \subset B \iff \text{pow}(A) \subset \text{pow}(B).$$

Hence, the given statement is TRUE.

Question 1.2.3 (Total 3.5 mark) (Rubric written at end.)

Statement: Let A, B be non-empty sets and let $f : A \to B$ be a function. For any subset $Z \subseteq A$, define:

$$F(Z) = \{f(a) \in B : a \in Z\}.$$

We must determine whether:

$$F(X \cap Y) = F(X) \cap F(Y)$$

is always true.

Definition of Function

We define a new function:

$$F : \text{pow}(A) \to \text{pow}(B)$$

by

$$F(Z) = \{f(a) : a \in Z\}.$$

- **Domain:** pow(A) — the power set of A
- **Codomain:** pow(B) — the power set of B

Step 1: Show one direction

Let $b \in F(X \cap Y)$. Then $\exists a \in X \cap Y$ such that $f(a) = b$. Since $a \in X$ and $a \in Y$, it follows that $b \in F(X)$ and $b \in F(Y)$. Therefore:

$$F(X \cap Y) \subseteq F(X) \cap F(Y).$$

Step 2: Check the reverse inclusion

Suppose $b \in F(X) \cap F(Y)$. Then there exist elements $a_1 \in X$ and $a_2 \in Y$ such that:

$$f(a_1) = b = f(a_2).$$

If $a_1 = a_2$, then $a_1 \in X \cap Y$ and thus $b \in F(X \cap Y)$. But if $a_1 \neq a_2$, then we cannot conclude that $b \in F(X \cap Y)$. Hence, equality may fail.

Step 3: Counterexample

Let:

$$A = \{1, 2\}, \quad B = \{x\}, \quad \text{and } f(1) = f(2) = x.$$

Take:

$$X = \{1\}, \quad Y = \{2\}.$$

Then:

$$F(X) = \{x\}, \quad F(Y) = \{x\} \implies F(X) \cap F(Y) = \{x\}.$$

However:

$$X \cap Y = \varnothing \implies F(X \cap Y) = \varnothing.$$

Therefore:

$$F(X \cap Y) \neq F(X) \cap F(Y).$$

Hence, the equality does **not always hold**.

Step 4: When it holds true

If f is an **injective (one-to-one)** function, then $f(a_1) = f(a_2)$ implies $a_1 = a_2$. In such a case, the previous issue cannot arise, and the equality

$$F(X \cap Y) = F(X) \cap F(Y)$$

will always hold.

Final Conclusion

The given statement:

$$F(X \cap Y) = F(X) \cap F(Y)$$

is **not true in general**. It holds **only when f is injective**.

$$\boxed{\text{Hence, the statement is FALSE in general.}}$$

Summary of Results

- **1.2.2:** The statement is **True**. Proper subset between sets implies and is implied by proper subset between their power sets.
- **1.2.3:** The statement is **False in general**, but becomes **True** if the function f is injective.

(+3.5) - will be provided if correct counter example is given to disprove.

Quiz3-Part2: Q1.3.2

Claim 4.1.3: For every set A, $\phi \subseteq A$.

Proof: Let A be an arbitrary set. We want to show $\phi \subseteq A$, which is equivalent to
$\forall x \in U (x \in \phi \Rightarrow x \in A)$, where U is the universal set.
As ϕ does not contain any element, $x \in \phi$ is false for every $x \in U$. Thus, from the truth table of $\Rightarrow$, we get that $\forall x \in U (x \in \phi \Rightarrow x \in A)$ is true. ■

Q1.3.1

Step 1: What is the domain and codomain of G∘F

$F : A \to B$

$G : B \to A$

So $G \circ F : A \to A$

This composition maps each element of A back to itself.

Step 2: Is G∘F injective?

To check injectivity, suppose $(G \circ F)(a_1) = (G \circ F)(a_2)$.

Then: $a_1 = a_2$ since $G \circ F = I$.

So G∘F is injective.

Step 3: Is G∘F surjective?

To check surjectivity, take any $a \in A$.

We must find some $x \in A$ such that $(G \circ F)(x) = a$.

But since $G \circ F = I$, $(G \circ F)(a) = a$.

So for each $a \in A$, $x = a$ works.

Therefore G∘F is surjective.

Step 4: Therefore…

Since $G \circ F = I$ is both injective and surjective, G∘F is bijective.

Therefore:

Function	Injective	Surjective	Bijective
$F : A \to B$	always	not always	Only if surjective
$G : B \to A$	not always	always	Only if injective
$G \circ F : A \to A$	yes	yes	Always bijective (since it equals I)

Can be Concluded using:

Lemma 4.4.2.2: let X, Y, Z be non-empty sets and $F : X \to Y$, $G : Y \to Z$ be functions. Then,

a) If F & G are both injective then $G \circ F$ is injective

b) If F and G are both surjective then $G \circ F$ is also surjective. (HW)

c) If F & G are both bijective then $G \circ F$ is also bijective.

d) If $G \circ F$ is injective then F is injective, but G need not be.

e) If $G \circ F$ is surjective then G is surjective, but F need not be.

Q1.3.3

$F : \mathbb{Q} \to \mathbb{Q}$, $F(x) = 2x - 1$

Step 1: Define Range and Codomain Codomain (B) is the set of all possible allowed outputs — here it's declared as Q (the rational numbers).

Range (or Image) is the set of actual outputs produced by F. By definition, Range(F) = $\{F(x) \mid x \in \mathbb{Q}\}$. Range(F) $\subseteq$ Codomain(F).

So it's always a subset — but we must check whether it's a proper subset (strictly smaller) or the whole codomain.

Step 2: Show what the Range actually is Let $y = F(x) = 2x - 1$.

Solve for x: $x = (y + 1)/2$. If $y \in Q$, then $x = (y + 1)/2 \in Q$ because rationals are closed under addition and division.
Hence every $y \in Q$ has a pre-image $x \in Q$. So Range(F) = Q
.
Step 3: The “subset” argument written formally Even though the range is a subset of the codomain by definition, Range(F) ⊆ Codomain(F), we’ve just shown they are equal here: Range(F) = Codomain(F) = Q.
Therefore, the range is not a proper subset — it’s the entire codomain.

Rubrics:

Q1.3.1
Correct Proof of why GoF is bijective: 2 marks
F(x) is always injective with justification: 1 marks
G(x) is always surjective with justification: 1 marks
Mentioning F(x) may or may not be Surjective: 1 marks
G(x) Injective
Q1.3.2
$\phi \subseteq A$ Equivalent to $\forall x \in U(x \in \phi \Rightarrow x \in A)$: 1 marks

$x \in \phi$ =False hence $\forall x \in U(x \in \phi \Rightarrow x \in A)$=True: 1 marks

Q1.3.3
Definition of range: $\{x+3 : x \in Z\}$ - 1 mark
Showing that for all $y \in Z$,$x = y - 3 \in Z$ - 1 mark
Concluding Range(F)=Z , so it’s not proper. - 1 mark